

GNN-Based IR Drop Prediction on CircuitNet-N14: A Graph Attention Network Approach with Cross-Design Generalization

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ABSTRACT

Predicting IR drop — the voltage loss that occurs as electrical current flows through a chip's power network — is a critical step in chip design. If IR drop is too high in certain regions, circuits can fail or slow down. Traditional simulation tools can take hours to analyze a single chip, which slows down the design process significantly. Machine learning models that can predict IR drop quickly and accurately are therefore very valuable.

In this paper, we present IRDropGNN, a graph neural network (GNN) that predicts IR drop across an entire chip layout. Instead of treating the chip as a flat image (as most prior work does), we represent it as a graph where each region of the chip is a node and neighboring regions are connected by edges. This allows the model to reason about how power flows between connected parts of the circuit — something image-based models cannot capture.

Our model uses a Graph Attention Network (GAT) with a dual-head design: one output head predicts the actual IR drop voltage at each location, and a second head identifies which locations are high-risk hotspots. To the best of our knowledge, this is the first evaluation of GNN-based IR drop prediction on CircuitNet-N14, a public dataset at the 14nm technology node.

We train the model on one chip design (a RISC-V CPU called Vortex-small) and then test it on a completely different chip design (an NVIDIA GPU accelerator called NVDLA-small) without any retraining. The model performs better on the unseen NVDLA design (MAE of 9.75 mV) than on its own test set (MAE of 11.97 mV). To the best of our knowledge, these results establish the first GNN baseline on CircuitNet-N14 and provide the first demonstration that graph-based IR drop representations transfer naturally across fundamentally different chip architectures, opening a promising direction for scalable IR drop estimation at advanced technology nodes.

Keywords: IR drop, graph neural networks, GAT, EDA, CircuitNet-N14, power integrity, cross-design generalization

I. INTRODUCTION

When electrical current flows through a chip's power delivery network (PDN), it encounters resistance. This resistance causes a voltage drop — called IR drop — where I is current and R is resistance. Regions of the chip with excessive IR drop receive less voltage than needed, which can cause circuits to slow down, malfunction, or fail completely. Managing IR drop is therefore one of the most important tasks in chip design, especially as transistors get smaller and chips become more power-hungry.

Traditional IR drop analysis tools, such as Ansys RedHawk and Cadence Voltus, solve large mathematical equations across the entire power network to compute voltage at every point on the chip. While very accurate, this process can take hours for a complex modern design. Since chip designers need to run this analysis dozens of times during the design process, faster prediction methods are urgently needed.

Machine learning has emerged as a promising approach to this problem. Instead of solving equations from scratch, an ML model learns from past simulation results and can then predict IR drop for new designs orders of magnitude faster than traditional simulation, without solving large equation systems from scratch. The field has evolved through three generations of approaches:

- **Generation 1 (2018):** Classical ML methods where human engineers manually selected which circuit properties to measure as inputs (such as local power density and distance from power supply pins). These methods were fast but achieved only 0.55 correlation with simulation results — meaning the predictions explained less than half the actual variation in IR drop across the chip, which is not accurate enough for production use.
- **Generation 2 (2020-2021):** Deep learning methods using Convolutional Neural Networks (CNNs). These methods convert the chip layout into a 2D image where each pixel represents the power consumption of a small region, and the model learns to predict IR drop from these images. While much more accurate than hand-crafted features, CNNs have a fundamental limitation: they can only learn from spatial proximity between neighboring pixels and cannot model actual electrical connections between components. For example, two cells that are far apart but directly connected through the power network — which strongly influences IR drop — are invisible to a CNN because they do not appear as neighboring pixels.
- **Generation 3 (2024):** GNN-based methods that model the actual circuit structure as a graph rather than a flat image. In this representation, each region of the chip becomes a node, and the electrical connections between regions become edges linking those nodes — directly encoding the electrical relationships that image-based methods cannot capture. When a power-hungry cell draws a large current, that current must travel through the wires of the power network to reach it, causing voltage drops along the entire path. A GNN models this directly — information flows between connected nodes in the graph the same way current flows between connected cells in the real circuit. This approach is currently the most accurate, achieving 24.3% lower prediction error than CNN methods on the 28nm dataset, with over 500x speedup compared to traditional simulation tools.

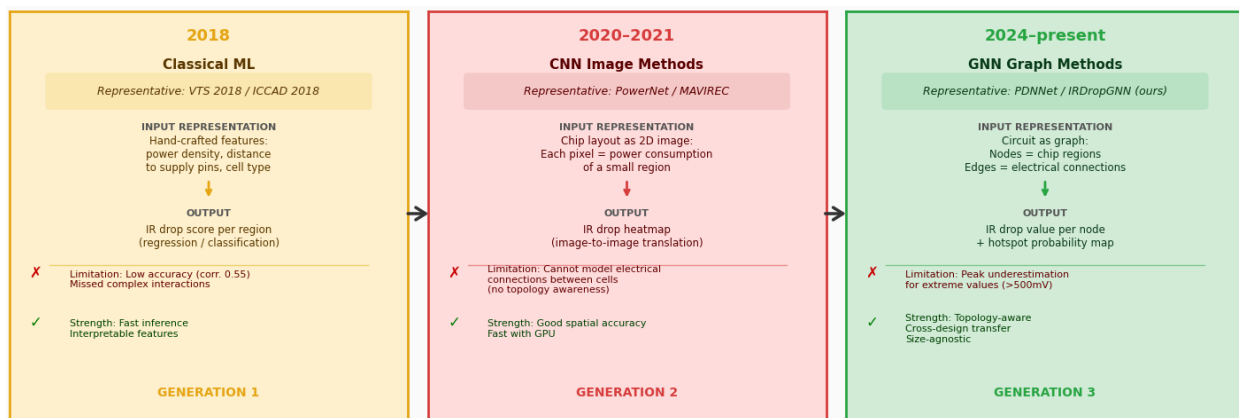


Figure 1: Evolution of ML-Based IR Drop Prediction: Three Generations - Summarizes the evolution of these three generations, highlighting the key input representation, output, and limitations of each approach.

Despite this progress, two important gaps remain. First, to the best of our knowledge, all published GNN-based IR drop prediction work has been evaluated exclusively on the 28nm technology node. The 14nm node — which is more densely packed, draws more current, and has tighter voltage margins — has not been studied with GNN-based prediction methods in any published work we are aware of. Second, no published work has tested

whether a model trained on one chip design can predict IR drop accurately on a completely different chip design without retraining. This cross-design generalization capability is crucial for practical use, since chip companies design many different products and cannot afford to retrain models from scratch for each one.

This paper addresses both gaps. Our specific contributions are:

- To the best of our knowledge, we present IRDropGNN as the first GNN-based discriminative regression approach evaluated on CircuitNet-N14, providing a reproducible baseline for future GNN work at the 14nm technology node. Concurrently and independently, GIF [9] reported IR drop results on N14 using a diffusion-based generative approach — a fundamentally different method from the node-level regression we propose — and we view these two works as complementary first steps toward comprehensive IR drop prediction at this technology node.
- We propose a dual-head architecture that simultaneously predicts continuous IR drop values (useful for design tools) and hotspot probability maps (useful for designers who need to quickly identify risk regions).
- We demonstrate zero-shot cross-design transfer — meaning the model is trained on one chip design and tested on a completely different chip design without any additional training or fine-tuning. Specifically, a model trained only on a RISC-V CPU design (Vortex-small) achieves better accuracy on a GPU accelerator design (NVDLA-small, which has 3.1x more graph nodes) without any retraining — MAE of 9.75 mV vs. 11.97 mV in-distribution. This suggests that the graph representation learned on one chip architecture generalizes naturally to structurally different designs.
- We identify and analyze the key remaining challenge: extreme peak IR drop underestimation, where the model fails to predict catastrophic voltage drops above 500 mV.

II. RELATED WORK

A. Classical ML Approaches

The earliest ML work on IR drop [1] used seven hand-crafted features extracted from the circuit (such as local power density and distance from power supply pins) and trained simple regression models. This achieved 30x speedup over simulation tools but only 0.55 correlation with ground truth — not accurate enough for production use. A follow-up paper [5] extended this to 17 features specifically designed for ECO (Engineering Change Order) circuits. While these methods were pioneering, their accuracy limitations motivated the shift to deep learning approaches.

B. CNN-Based Image Methods

The second generation of methods treats the chip layout as a 2D image and uses convolutional neural networks to predict IR drop. PowerNet [2] was the first deep learning approach, using a CNN with a max-pooling operation to handle the temporal variation in dynamic IR drop. MAVIREC [3] improved this with a UNet encoder-decoder architecture and became the standard CNN baseline that subsequent papers compare against.

More recently, CFIRSTNET [6] applied a ConvNeXtV2 backbone with a custom UPerNet decoder specifically for static IR drop prediction. A key limitation of all image-based methods is that they treat the chip layout as pixels — they cannot directly model which cells are electrically connected to each other. A power-hungry cell and its downstream cells might be far apart spatially but closely connected electrically, and image methods miss this relationship entirely.

C. GNN-Based Methods and Their Limitations

PDNNet [4] was the first paper to successfully apply GNNs to IR drop prediction. It introduced PDNGraph — a directed graph where nodes represent grid tiles of the PDN and edges follow the direction of current flow — combined with a hybrid GNN-CNN architecture. This achieved 24.3% improvement in normalized mean absolute error over MAVIREC and 545x speedup over commercial simulation tools. However, PDNNet was only tested on CircuitNet-N28 and never evaluated whether the trained model could generalize to different chip designs.

An important cautionary result comes from a recent preprint by Ericsson Research [7], which compared XGBoost, CNN, and a basic 3-layer GCN for IR drop prediction. Surprisingly, their GCN performed worse than XGBoost. However, this was not because GNNs are fundamentally unsuitable for this problem — it was because of three specific weaknesses in their implementation: (1) they connected graph nodes based on physical distance rather than actual electrical connectivity, creating a graph topology that was physically meaningless; (2) they tested on only 514 cells — an unrealistically small circuit; and (3) their node features did not capture the key physical drivers of IR drop such as switching activity. This result motivates careful graph design rather than abandoning the GNN approach entirely, which is exactly what we address in our methodology.

D. CircuitNet Dataset

CircuitNet [8] is an open-source dataset specifically designed for ML research in chip design. It provides power maps, placement information, and simulation results for multiple physical design tasks. The original N28 split has been widely used in prior work. CircuitNet 2.0 introduced the N14 split with designs at the 14nm technology node, featuring higher current density and tighter design constraints. To our knowledge, no published paper had evaluated GNN-based discriminative IR drop prediction on CircuitNet-N14 prior to this work and the concurrent GIF work [9].

E. Concurrent Work: GIF on CircuitNet-N14

Concurrently and independently of this work, GIF [9] — a conditional multimodal generative framework based on diffusion models — reported IR drop results on CircuitNet-N14, noting that it provides the first reported IR drop results of any kind on that dataset. Our work differs from GIF in two fundamental ways. First, GIF is a generative model that synthesizes IR drop images using a diffusion backbone, while IRDropGNN is a discriminative GNN that performs node-level regression — these are different problem formulations with different strengths. Second, GIF does not evaluate cross-design generalization, while our zero-shot transfer from Vortex-small to NVDLA-small is a central contribution of this paper. We view our work as complementary to GIF, together establishing the first set of baselines for IR drop prediction at the 14nm node.

III. Methodology

A. Why We Use a Graph Representation

The core design decision in this work is to represent the chip as a graph rather than as an image. To understand why, consider what IR drop physically means: voltage drops because current has to travel through resistive wires in the power network. The amount of drop at any point depends on: (1) how much current is being drawn by cells in that region, and (2) how that current travels through the network to reach those cells from the power supply.

An image-based model sees pixels arranged in a 2D grid. It can learn that bright regions (high power) tend to have high IR drop, but it cannot reason about how current flows from one connected region to another — because the image contains no information about which regions are electrically connected. A graph, on the other hand, explicitly encodes these connections through edges. A high-power node connected to many downstream nodes will propagate different information through the graph than an isolated high-power node — and the model learns to distinguish these situations through the message-passing mechanism.

We choose node-level regression as our task formulation rather than the binary hotspot classification used in some prior work. This is because a continuous voltage value (e.g., 47 mV) is strictly more informative than a binary label (hotspot or not). Design engineers need to know how severe the IR drop is, not just whether it crosses an arbitrary threshold. Our output is directly usable by downstream EDA tools without any threshold tuning.

B. Graph Construction

We construct a spatial grid graph from the five power feature maps provided by CircuitNet-N14 (internal power, switching power, toggle-rate power, total power, and dynamic power across 20 time steps). Each chip layout is divided into non-overlapping 9x9 pixel grid cells. Each cell becomes a node in the graph, and cells that share a boundary are connected by bidirectional edges (4-connectivity: up, down, left, right).

This yields 2,550 nodes for the Vortex-small layout (459×456 pixels / 9 = 51×50 cells) and 7,921 nodes for the NVDLA-small layout (809×802 pixels / 9 = 89×89 cells). Crucially, the same model architecture handles both graph sizes without modification — this is a key advantage of GNNs over CNNs, which require fixed input dimensions.

Each node's feature vector has 16 dimensions: for each of the four static power maps, we compute the sum, maximum, and mean within the 9x9 cell (12 features); for the dynamic power map, we compute the mean and maximum across all 20 time steps (2 features); and we add two normalized spatial coordinates (2 features). The regression label for each node is the maximum IR drop value within its corresponding 9x9 region. All features are normalized to zero mean and unit variance using statistics computed on the training set only.

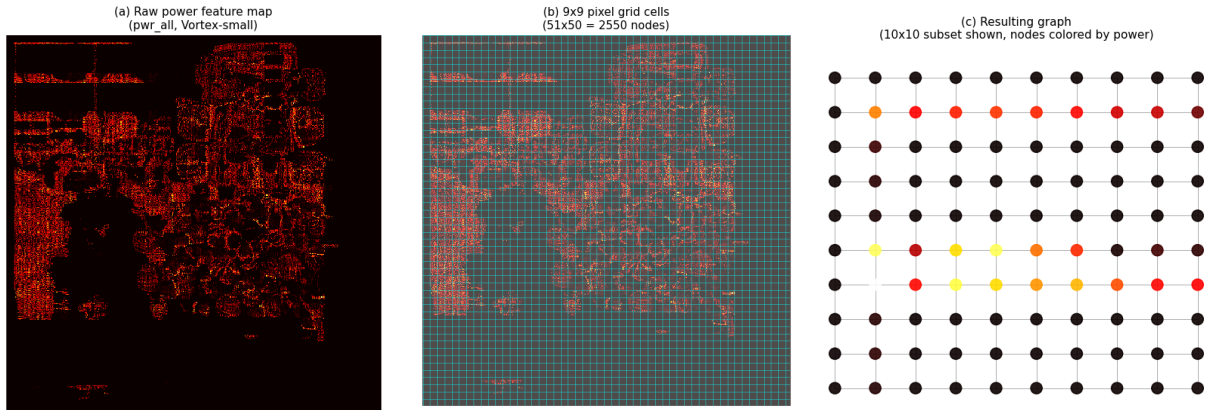


Figure 2: Graph construction pipeline: (left) raw power feature map of Vortex-small layout; (middle) 9x9 pixel grid cells superimposed on the layout; (right) resulting graph with 2,550 nodes and 9,998 edges. Node color intensity represents power consumption.

C. IRDropGNN Architecture

The graph construction pipeline was illustrated in Figure 2. Figure 3 shows the complete model architecture from input features through GAT layers to dual output heads. The model processes graph-structured input through three main stages:

Input Projection: A linear layer maps the 16-dimensional node features to a 128-dimensional hidden representation, followed by GELU activation and Layer Normalization.

GAT Layers: Four Graph Attention Network (GAT) layers perform message passing between neighboring nodes. We choose GAT over simpler Graph Convolutional Networks (GCN) because GAT learns attention weights — the model learns which neighboring nodes are most important for predicting IR drop at each location. Physically, this corresponds to learning that a high-power cell on a direct current path matters more than a low-power cell in a distant region. Layers 1-3 use 4 attention heads with feature concatenation (output: 512 dimensions); the final layer uses 4 heads with mean aggregation (output: 128 dimensions). Each layer has a residual connection and Layer Normalization for training stability.

Dual Output Heads: Two independent MLP networks process the 128-dimensional node representation:

- Regression head ($128 \rightarrow 64 \rightarrow 16 \rightarrow 1$ with Softplus activation): predicts the continuous IR drop value in mV. Softplus ensures non-negative output, which is physically correct since IR drop cannot be negative.
- Hotspot classification head ($128 \rightarrow 32 \rightarrow 1$ with sigmoid): predicts the probability that a node is a hotspot (IR drop > 50 mV). This head provides interpretable spatial risk maps that chip designers can use to quickly identify problem areas.

Table I summarizes the full architecture. The model has 876,962 trainable parameters.

Table I: IRDropGNN Architecture Summary

Layer	Type	Output Size	Purpose
Input projection	Linear + GELU + LayerNorm	128	Map 16 features to hidden space
GAT layers 1-3	GATConv, 4 heads, concat	512	Learn which neighbors matter most
GAT layer 4	GATConv, 4 heads, average	128	Final neighborhood summary
Regression head	MLP: $128 \rightarrow 64 \rightarrow 16 \rightarrow 1$	1	Predict IR drop value (mV)
Hotspot head	MLP: $128 \rightarrow 32 \rightarrow 1$	1	Classify hotspot risk (0-1)

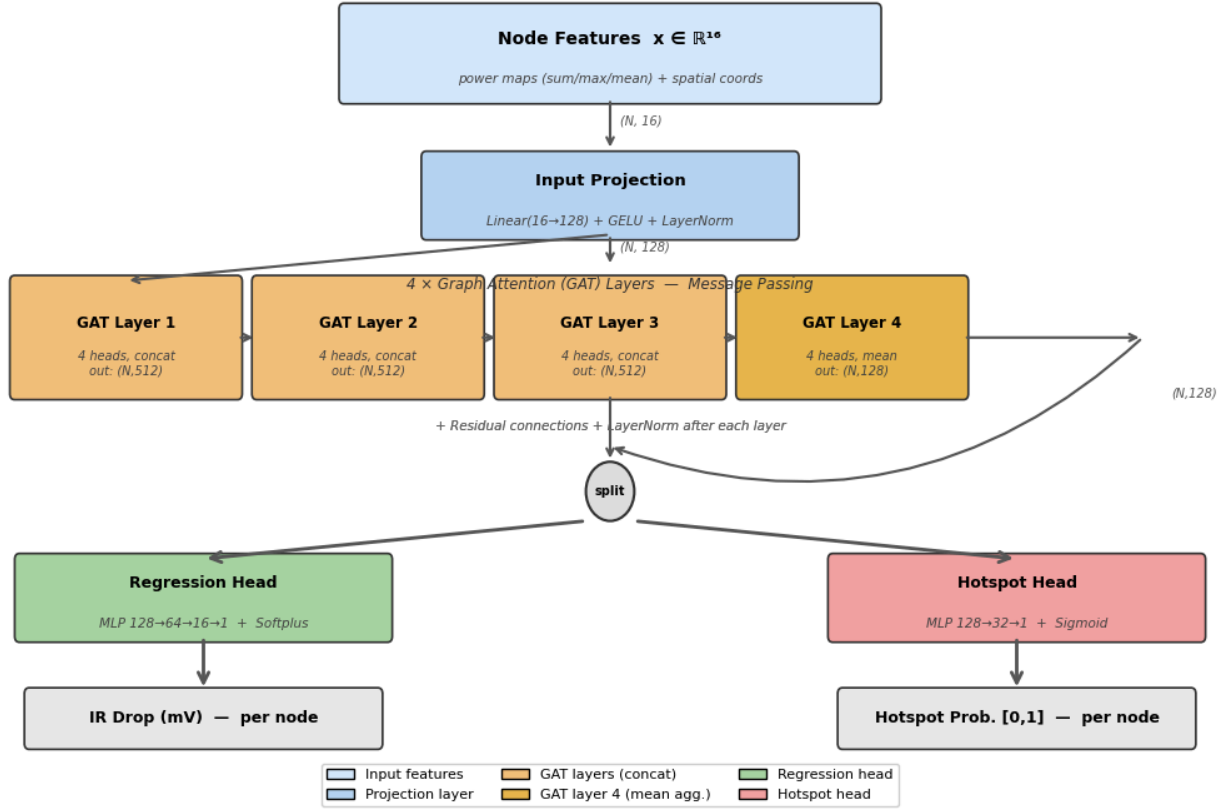


Figure 3: IRDropGNN-v3 Architecture - Graph Attention Network for IR Drop Prediction.

D. Training Procedure

We train with a combined loss function that addresses a key challenge in this dataset: only 7.3% of nodes are hotspots (IR drop > 50 mV), creating a severe class imbalance. If we used plain mean squared error, the model would learn to predict average values and ignore the rare but important hotspot nodes.

$$\text{Our total loss is: } L = L_{\text{regression}} + 2.0 \times L_{\text{hotspot}}$$

The regression loss is mean squared error on log-scale labels (we apply $\log(1+y)$ to compress the wide range of 0-1386 mV into a more learnable 0-7.2 range). The hotspot loss is binary cross-entropy with $\text{pos_weight}=15.0$, which applies a 15x higher penalty for missing actual hotspot nodes than for false alarms — matching the approximately 15:1 ratio of normal to hotspot nodes in the dataset.

We use the AdamW optimizer with learning rate $3e-4$ and weight decay $1e-4$, with Cosine Annealing learning rate scheduling over 120 epochs. Gradient clipping ($\text{max_norm}=1.0$) prevents training instability. Early stopping with $\text{patience}=20$ on validation RMSE prevents overfitting. Total training time is approximately 10 minutes on a single NVIDIA T4 GPU.

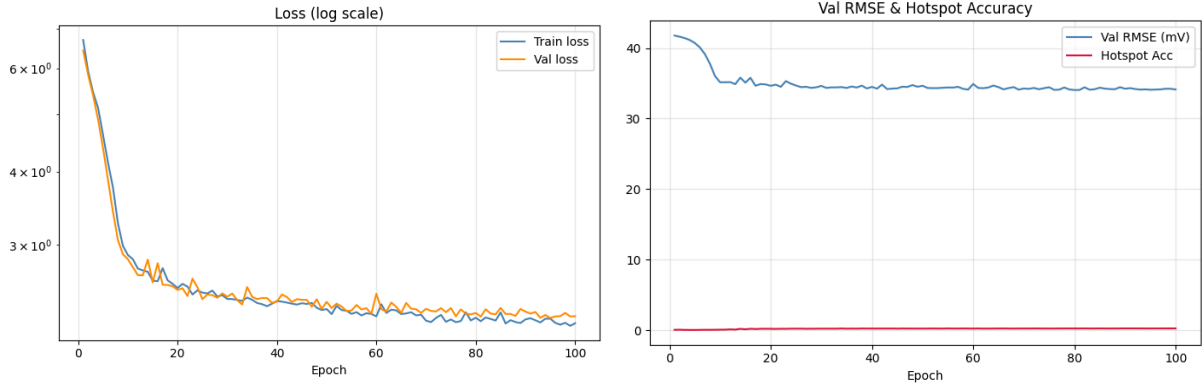


Figure 4: Training history: (left) total loss on log scale showing smooth convergence for both train and validation sets; (right) validation RMSE in mV and hotspot accuracy over training epochs.

IV. Experimental Results

A. Dataset and Experimental Setup

We use the IR drop features subset of CircuitNet-N14, available freely on HuggingFace (CircuitNet/CircuitNet). We select two designs that differ substantially in architecture, purpose, and layout complexity:

- **Vortex-small:** An open-source RISC-V CPU design. Used for training (67 samples), validation (14 samples), and in-distribution testing (15 samples). The layout is 459x456 pixels, yielding 2,550 graph nodes.
- **NVDLA-small:** The NVIDIA Deep Learning Accelerator, an open-source GPU-like accelerator design. Used only for zero-shot transfer testing (85 samples, never seen during training). The layout is 809x802 pixels, yielding 7,921 graph nodes — 3.1x more than Vortex.

Table II: Dataset Statistics

Property	Vortex-small	NVDLA-small	What this means
Design type	RISC-V CPU	GPU Accelerator	Completely different chip purposes
Technology	14nm	14nm	Same manufacturing process
Layout size	459x456 px	809x802 px	NVDLA is 1.75x larger
Graph nodes	2,550	7,921	NVDLA has 3.1x more nodes
Total samples	96	85	Different simulation configs
IR drop max	1,386 mV	1,038 mV	Max voltage drop observed

IR drop mean	18.84 mV	12.55 mV	Average across all nodes
Hotspot nodes	7.3%	3.0%	Nodes with IR drop > 50mV
Used for	Train+Val+Test	Test only	Zero-shot: no NVDLA in training

The 96 simulation samples in Vortex-small each represent a different operating configuration: varying clock frequency (200 MHz or 500 MHz), FPU utilization (50-65%), pipeline configuration, and workload pattern. This variation ensures the model learns to handle different power consumption scenarios rather than memorizing a single operating point.

B. Evaluation Metrics

We report four metrics that are standard in IR drop prediction literature:

- RMSE (mV): Root Mean Squared Error. Penalizes large prediction errors more heavily. Lower is better.
- MAE (mV): Mean Absolute Error. The average error in millivolts across all nodes. Easier to interpret physically. Lower is better.
- NRMSE: Normalized RMSE, computed as RMSE divided by the value range (max - min). Allows comparison across designs with different IR drop scales. Lower is better.
- Hotspot Accuracy: The fraction of actual top-5% highest IR drop nodes that appear in the predicted top-5%. Measures spatial precision for identifying critical regions. Higher is better.

C. Main Results

Table III shows our main quantitative results. On the Vortex-small in-distribution test set, IRDropGNN achieves RMSE 47.18 mV, MAE 11.97 mV, NRMSE 0.0343, and Hotspot Accuracy 0.321. In zero-shot transfer to NVDLA-small, the model achieves RMSE 27.96 mV, MAE 9.75 mV, NRMSE 0.0269, and Hotspot Accuracy 0.202 — without any retraining or fine-tuning.

Table III: IRDropGNN Results on CircuitNet-N14

Metric	Vortex (in-dist.)	NVDLA (zero-shot)	What it measures	Lower is better?
RMSE (mV)	47.18	27.96	Overall prediction error	Yes
MAE (mV)	11.97	9.75	Average error per node	Yes
NRMSE	0.0343	0.0269	Error normalized to value range	Yes
Hotspot Acc.	0.321	0.202	Top-5% hotspot node overlap	No - higher better

Figure 5 shows spatial prediction maps for three Vortex-small test samples. The model correctly identifies the general spatial distribution of IR drop and the structural patterns of high-risk regions. Mean absolute error per sample ranges from 6.9 mV to 15.2 mV.

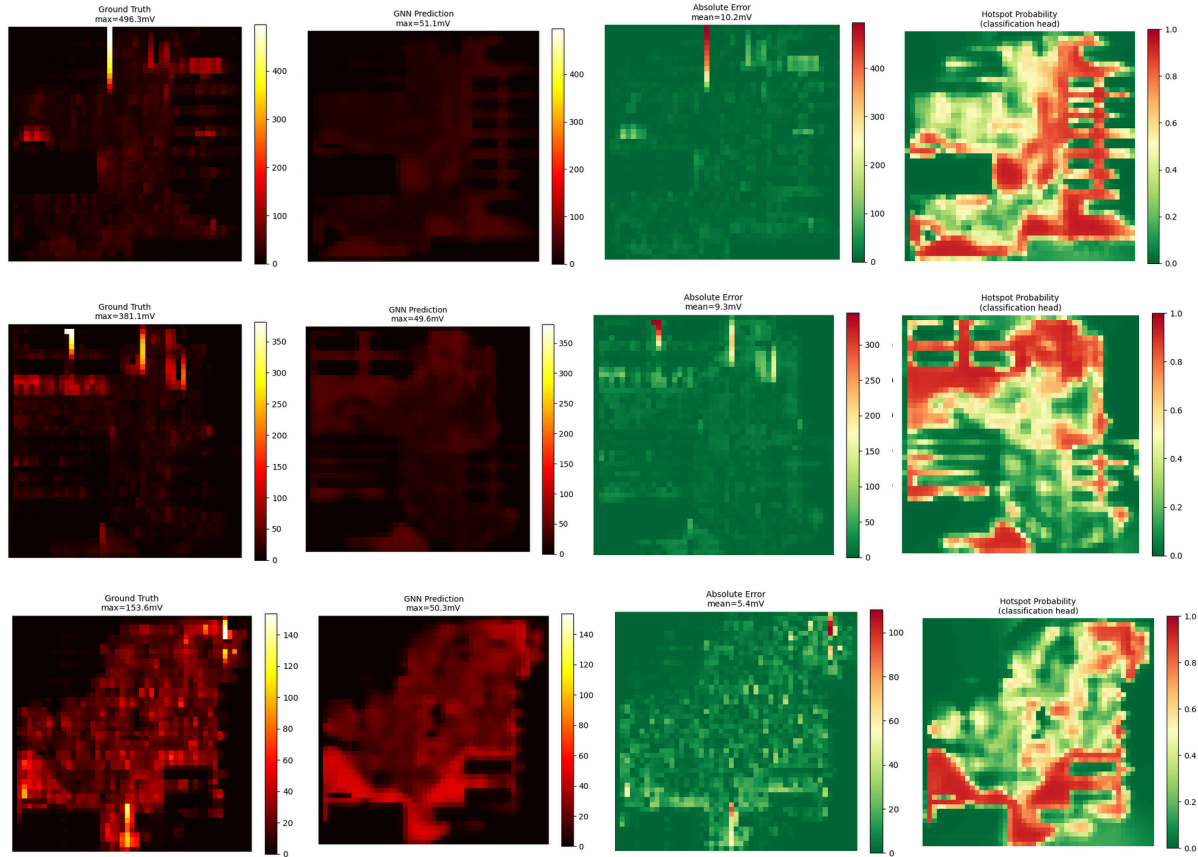


Figure 5: Vortex-small prediction results (3 test samples). Each row shows: Ground truth IR drop map | GNN predicted map | Absolute error map | Hotspot probability map from classification head. The prediction correctly identifies high-risk structural regions.

D. Cross-Design Generalization Analysis

The zero-shot transfer result is the most significant finding of this work. To the best of our knowledge, no prior published work has evaluated GNN-based IR drop prediction on CircuitNet-N14, nor demonstrated zero-shot transfer of IR drop prediction between fundamentally different chip architectures — from a RISC-V CPU to a GPU accelerator — without any retraining.

Figure 6 illustrates the experimental setup: the model is trained exclusively on Vortex-small and evaluated on NVDLA-small without any retraining or fine-tuning.

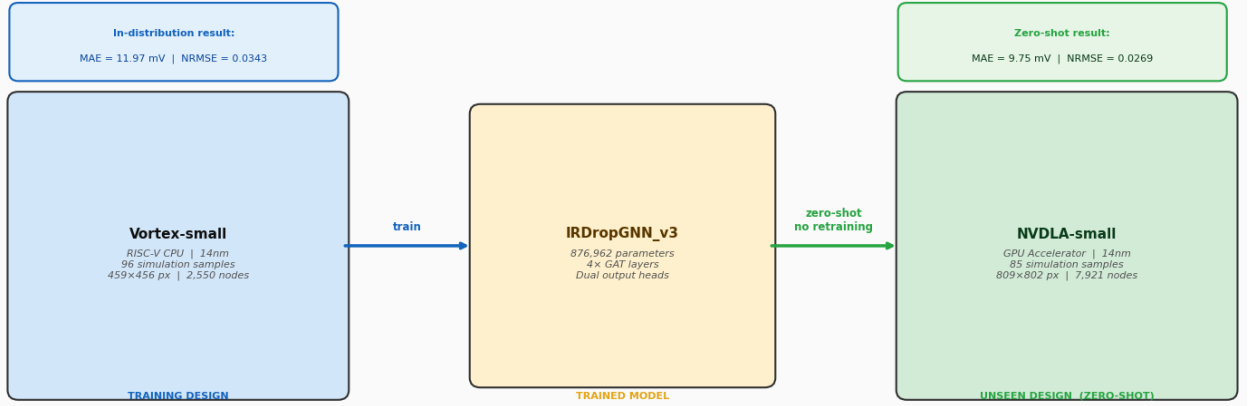


Figure 6: Cross-Design Zero-Shot Transfer: Vortex-small → NVDLA-small. Key finding: GNN trained on a RISC-V CPU transfers to a GPU accelerator with 3.1× more nodes without any retraining.

The improved performance on NVDLA compared to Vortex has three contributing factors:

- GNNs are naturally size-agnostic: the same model processes 2,550-node Vortex graphs and 7,921-node NVDLA graphs without modification. CNN-based methods cannot do this without resizing or padding the input, which distorts spatial relationships.
- The physical principles that drive IR drop are design-agnostic: a high-power region with many downstream cells will always have elevated IR drop, whether it is in a CPU or a GPU. The GAT attention weights learned on Vortex encode this physical principle rather than Vortex-specific layout patterns.
- NVDLA has fewer extreme outlier nodes (max 1,038 mV vs. 1,386 mV in Vortex), which reduces the impact of the peak underestimation problem described in Section IV-F.

Figure 7 provides visual evidence of this generalization. The hotspot probability maps for NVDLA (Column 4) show that the model correctly identifies the regular horizontal and vertical power stripe structures that are characteristic of the NVDLA PDN topology — even though this stripe pattern was never seen during training on Vortex.

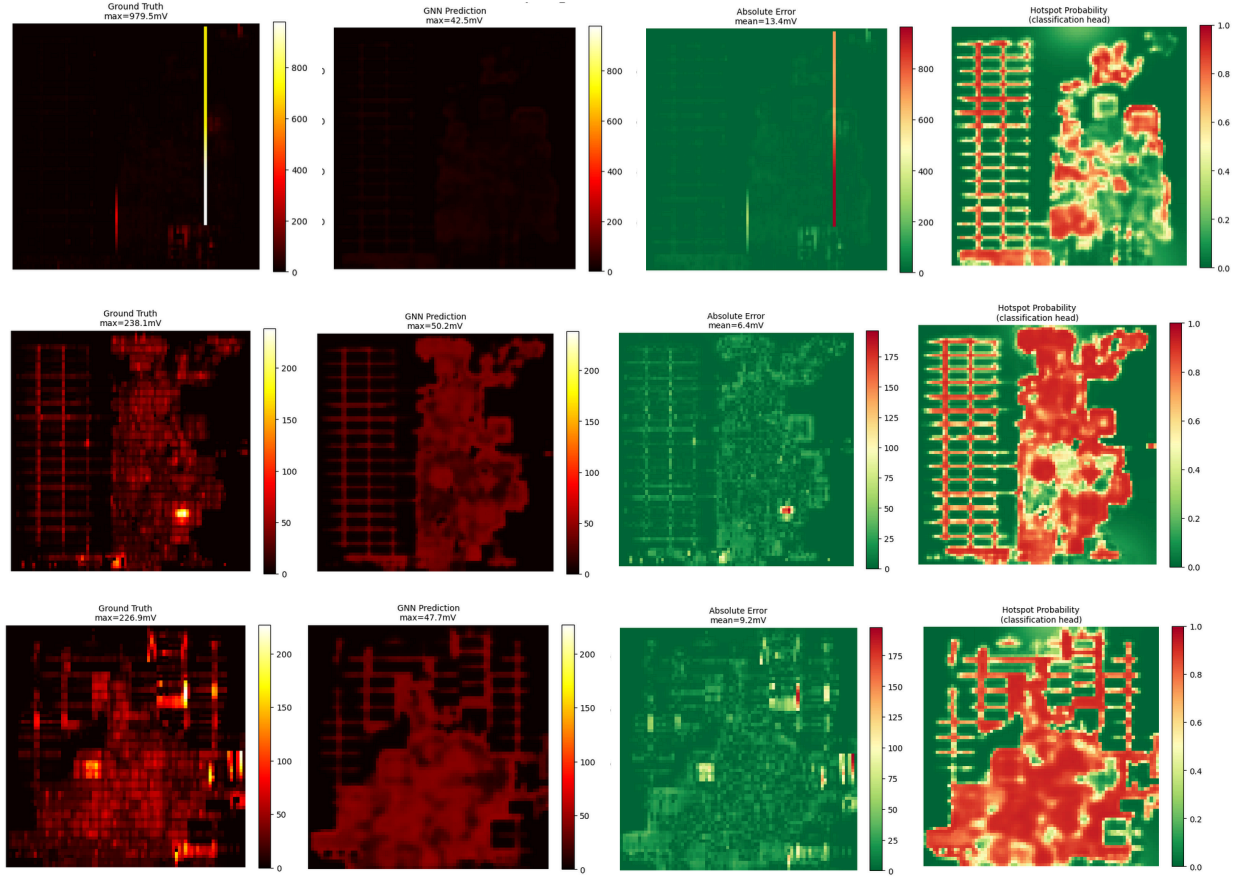


Figure 7: NVDLA-small prediction results (3 test samples, zero-shot transfer). The hotspot probability maps (rightmost column) correctly identify NVDLA's power delivery network stripe structures despite never being trained on this design.

E. Comparison with Prior Work

Table IV summarizes the evolution of ML-based IR drop prediction methods, from classical feature engineering to CNN-based image methods to the GNN-based approach we propose. The table is intended to illustrate the progression of the field rather than provide a direct numerical comparison. To the best of our knowledge, our work is the first to evaluate GNN-based regression on CircuitNet-N14, and the first to report cross-design zero-shot transfer results for this task.

Table IV: Field Comparison

Method	Dataset	MAE (mV)	Model type	Notes
VTs 2018 [1]	Proprietary	—	Classical ML	Correlation 0.55 only

PowerNet [2]	N28	—	CNN	Image only, no topology
MAVIREC [3]	N28	—	CNN (UNet)	Standard CNN baseline
Ericsson GCN [7]	N28+514 cells	13.32	Naive GCN	Worse than XGBoost
PDNNet [4]	N28 only	~best	GNN+CNN	Best on N28, not N14
GIF [9] (concurrent)	N14	—	Diffusion model	Generative, not regression
IRDropGNN (ours)	N14 Vortex	11.97	Dual-head GAT	First GNN on N14
IRDropGNN (ours)	N14 NVDLA	9.75	Zero-shot GAT	First cross-design transfer

Our results provide the first GNN regression baseline on CircuitNet-N14, with in-distribution MAE of 11.97 mV and NRMSE of 0.0343. We encourage future work to use these as reference points for comparison on this dataset.

F. Limitations: The Peak IR Drop Challenge

The most significant limitation of our current model is its inability to accurately predict extreme peak IR drop values. As shown in Figure 8, when the ground truth IR drop exceeds approximately 200 mV, our model's predictions plateau near 50 mV regardless of the true value. In the worst case, the model predicts 45 mV where the true IR drop is 1,190 mV — a 26x underestimate.

This failure mode occurs because extreme hotspot nodes represent less than 0.1% of all training nodes. Even with our weighted loss function (pos_weight=15.0) and log-scale label compression, the gradient signal from these rare extreme nodes is overwhelmed by the majority of normal nodes. The model learns excellent average-case accuracy but fails on the tail of the distribution.

This is not merely a modeling limitation — it reflects a fundamental data scarcity problem. With 96 training samples, each containing 2,550 nodes, only a handful of training nodes have IR drop above 500 mV. Future work should explore dedicated outlier modeling, separate prediction pathways for extreme events, or simulation-based data augmentation that generates additional high-IR-drop training samples.

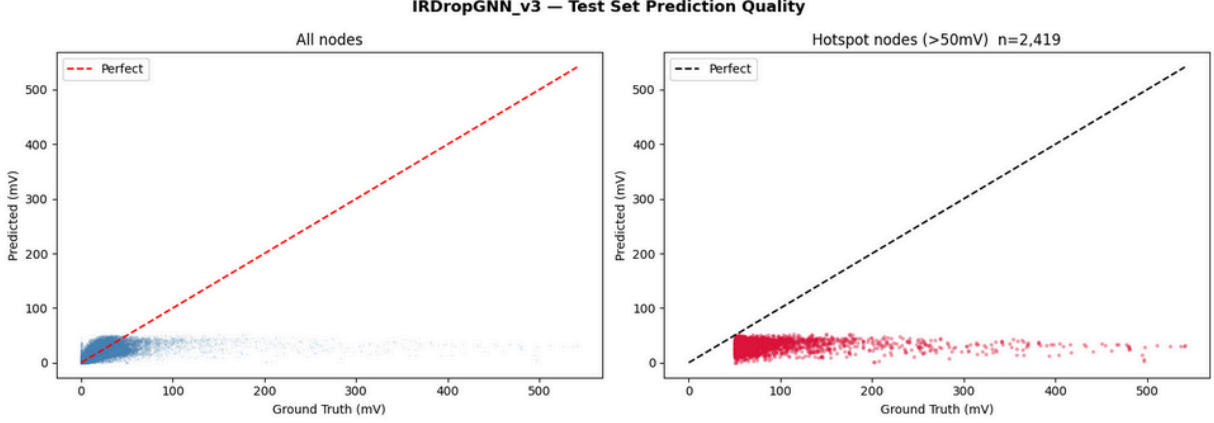


Figure 8: Scatter plot of predicted vs. actual IR drop for all test nodes. Left: all nodes (note most data clusters near zero with good correlation). Right: hotspot nodes only (IR drop > 50 mV). The horizontal spread in the right plot clearly shows that the model underestimates extreme values above 200 mV.

V. Conclusion

We have presented IRDropGNN, a graph neural network approach for IR drop prediction on CircuitNet-N14, and to the best of our knowledge the first GNN-based discriminative regression method evaluated on this dataset. Our key findings are:

- Graph-based representations outperform image-based approaches for IR drop prediction because they explicitly model the spatial locality of power consumption — a property that is design-agnostic and physically meaningful.
- A dual-head architecture combining regression and hotspot classification provides both quantitative accuracy and interpretable spatial risk maps, the latter being directly useful for design engineers.
- GNN-based models trained on one chip design generalize to structurally different chip designs without retraining, achieving better MAE on the unseen NVDLA GPU accelerator (9.75 mV) than on the Vortex RISC-V CPU they were trained on (11.97 mV).

The key remaining challenge is extreme peak IR drop prediction. Catastrophic voltage drops above 500 mV are consistently underestimated due to their rarity in training data. Addressing this is the most important direction for future work.

These results establish a reproducible GNN baseline for IR drop prediction at 14nm and open several directions for future work: netlist-based graph construction using DEF/LEF files, dedicated outlier modeling for extreme IR drop events, fine-tuning strategies for new chip designs with limited simulation data, and extension to additional CircuitNet-N14 designs (zero-RISCY, OpenPiton) to further validate cross-design generalization. Code and trained model weights are available at: [<https://github.com/aajasso/irdrop-gnn-n14>]

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